

AGNIBH DASGUPTA

♥ Omaha, NE ✉ adg002@gmail.com ☎ +1 (435) 754-9599 🌐 [Website](#) 🔗 [Linkedin](#) 🐙 [Github](#) 🎓 [GScholar](#)

Education

University of Nebraska Omaha, Ph.D., Computing & Information Science	2021 – 2026
Utah State University, M.S., Computer Science	2018 – 2020

Skills

Research:	Representation Learning, Watermarking, Computer Vision, NLP, SSL, Multimodality
Models:	Transformers, ViT, CLIP, LLMs, Diffusion Models
ML Stack:	Python, PyTorch, TensorFlow, Hugging Face Transformers, CUDA
Training:	LoRA, SFT, DPO, GRPO, RLHF, RLAIF, PyTorch DDP, DeepSpeed, FSDP, Slurm/HPC
Agentic AI:	RAG, VectorDB, LangGraph, Tool Use, Function Calling
Evaluation:	LLM-as-a-Judge, RAGAS, A/B Testing, Human-in-the-Loop
Deployment:	vLLM, Quantization, Cloud Deployment, Docker

Professional Experience

University of Nebraska Omaha & University of Nebraska Medical Center, Postdoc	2026 – Present
Conducting applied AI research on multimodal datasets at the Munroe-Meyer Institute to better understand and support individuals with disabilities.	
University of Nebraska Omaha, Graduate Researcher	2021 – 2026
- Led papers and experiments, hiring, interviewing, and mentoring interns in weekly meetings - Worked in interdisciplinary teams with education researchers at UNL and doctors at Stanford - Served as a peer reviewer, presented at conferences, authored NSF and graduate grant proposals	

Awards

CVPR Compute Transparency Champion Award	2026
Awarded for exemplary reporting of computational resources, methodology, and reproducibility details	

Selected Publications and Projects

Siosa's Library: Path of Exile Q&A Agent	2026
--	------

- Built a RAG based tool using **agentic AI** workflows to help new players navigate the game's difficulty
- Designed a **multi-agent** workflow using **LangGraph** to coordinate and tool selection across multiple data sources
- Hosted a pretrained LLM on **cloud infrastructure**, optimizing inference with **vLLM** and **quantization**
- Aligned model behavior via **human-in-the-loop post-training (DPO, PPO, GRPO)**, evaluated with **LLM-as-judge**

Invariant Features in LLMs: Geometric Characterization and Model Attribution	NeurIPS 2026
--	--------------

- Investigated whether LLMs encode causally meaningful **representations** that can be used for **model attribution**
- Identified **invariant subspaces** in 9 open-source LLMs (7-10B params) via contrastive sensitivity decomposition
- Validated semantic and nuisance components causally through next-token prediction interventions
- Achieved 90%+ **zero-shot model attribution** accuracy across base, **fine-tuned**, and **distilled LLMs**

TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation	CVPR 2026
---	-----------

- Addressed the challenge of maintaining **watermark** robustness under real-world camera capture conditions
- Built a **zero-watermarking** framework combining **CLIP** image-text features with a learned **augmentation** module
- Trained a differentiable auto-augmentor to learn camera-like distortions, replacing hand-designed **augmentation**
- Achieved 95%+ watermark recovery on **ImageNet** and **Visual Genome** under real-world camera conditions

Multimodal Sentiment Analysis in K-12 Classrooms (in progress)	2026
--	------

- Extending a published **video-only** classroom mood classification pipeline (IEEE FIE 2024) lacking audio
- Collaborating with education researchers at **UNL** to build a **multimodal** system for richer instructor feedback
- Combined **video** and **audio** features (**log-mel spectrograms**, acoustic uncertainty) to model student engagement
- Ongoing: evaluating whether the added audio modality improves on the published video-only baseline

Watermarking Language Models through Language Models	IEEE TAI 2025
--	---------------

- Addressed the need for **LLM watermarking** that generalizes across architectures without modifying model weights
- Built a model-agnostic, prompt-guided **watermarking** scheme for **LLM-generated** text
- Tested on 25 open-source LLMs including **fine-tuned** and **distilled** variants using **LoRA** and **DeepSpeed**
- Achieved 88%+ detection accuracy under paraphrasing, **fine-tuning**, and **adversarial prompt attacks**

Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning	CSCI 2023
--	-----------

- Addressed the need of a **watermarking** scheme robust to real-world noise via a learned representation
- Built a **ViT-based cross-attention watermarking** system with a **self-supervised** triplet loss objective

- Trained the encoder to cluster augmented views while separating different images in latent space
- Achieved 90%+ **bit recovery rate (BRR)**, comparable to prior state-of-the-art under compound noise

Perspective Transformation Layer

CSCI 2022

- Addressed the limitation of fixed preprocessing pipelines for handling viewpoint variation in image recognition
- Proposed a lightweight learnable layer for perspective-robust recognition, replacing fixed pipelines
- Evaluated on **SVHN**, Imagenette, and **MNIST** under viewpoint variation with minimal extra compute
- Achieved 96%+ classification accuracy, outperforming prior state-of-the-art on perspective benchmarks

Deep Q-Learning Applied to Stock Trading

M.S. Thesis, 2020

- Designed a **reinforcement learning** agent to **predict** next-day **S&P 500** price movement from historical index data
- Represented price history as **candlestick charts** to emulate human trader pattern recognition
- Automated daily scraping, **prediction**, and **model** fine-tuning into one self-updating pipeline
- Observed positive portfolio returns from the agent's **prescribed** buy/sell/hold actions